Contents

Summary

Chapter 1 Systems of Linear Equations **2**

1.1 Systems of Linear Equations 3

1.1.1 Forming the Equations 3

1.1.2 Special Case: Number of Equations > Number of Variables 3

1.1.3 Special Case: Number of Equations < Number of Variables 4

Chapter 2 Inequalities **5**

2.1 Solving Inequalities Graphically 6

2.2 Sign Test Technique 6

2.3 Common Substitutions 7

Summary 1

Systems of Linear Equations

1.1 Systems of Linear Equations

1.1.1 Forming the Equations

Example

$$\left.\begin{aligned} a+b+c&=176 && —(1)\\ 0.85a+0.90b+0.95c&=161.80 && —(2)\\ 0.93a+0.95b+0.98c&=169.56 && —(3)\end{aligned}\right\} \Rightarrow a=32,\ b=44,\ c=100$$

Example

The "variables" may be expressions. To find x, y and z, use the given information to form the equations:

$$\left.\begin{aligned} 2x^3+3\ln y-e^z&=18 && —(1)\\ x^3-\ln y+2e^z&=9 && —(2)\\ x^3+2\ln y+e^z&=11 && —(3)\end{aligned}\right\} \Rightarrow A=8,\ B=1,\ C=1$$

It is technically still 'linear' because we can think of $A=x^3$, $B=\ln y$, $C=e^z$, which turns it into a linear system in A, B and C. A truly nonlinear equation could look like $x^2+x+y^2+2y-3z^3-xz+xy=1$; it cannot be made linear this way, because x appears as both x^2 and x, and there are product terms such as xz and xy.

1.1.2 Special Case: Number of Equations > Number of Variables

Example

$$\left.\begin{aligned} \lambda+\mu&=1 && —(1)\\ 2\lambda-\mu&=5 && —(2)\end{aligned}\right\} \Rightarrow \lambda=2,\ \mu=-1$$

$$\lambda+3\mu=-1 \qquad —(3)$$

Sometimes there may be no solution, e.g. skew lines in vectors. In this case, solve for λ and μ using equations (1) and (2). Then sub into (3).

> **Note**
>
> Check (3): if LHS = RHS, the equations are consistent; if LHS ≠ RHS, there is no solution.

$$\begin{aligned}\text{Sub into (3): LHS}&=2+3(-1)\\ &=-1\\ &=\text{RHS}\\ \therefore \lambda&=2,\ \mu=-1\end{aligned}$$

1.1.3 Special Case: Number of Equations < Number of Variables

Example

$$\left.\begin{aligned} x+y+z=6 \quad &—(1) \\ x-y+2z=5 \quad &—(2) \end{aligned}\right\} \Rightarrow x=\tfrac{11}{2}-\tfrac{3}{2}z,\ y=\tfrac{1}{2}+\tfrac{1}{2}z,\ z=z$$

> **Note**
>
> Useful trick: in contextual questions, the variables are usually non-negative integers, $z \in \mathbb{Z}_0^+$, so use this to find the solution from the general solution.

When $z=z$, it means that there are infinitely many solutions, and z can be any real value. So just sub that into x and y to get a correct solution.

$$z=-1 \Rightarrow (x,y,z)=(7,\ 0,\ -1)$$
$$z=0 \Rightarrow (x,y,z)=(\tfrac{11}{2},\ \tfrac{1}{2},\ 0)$$
$$z=1 \Rightarrow (x,y,z)=(4,\ 1,\ 1)$$
$$z=2 \Rightarrow (x,y,z)=(\tfrac{5}{2},\ \tfrac{3}{2},\ 2)$$
$$z=3 \Rightarrow (x,y,z)=(1,\ 2,\ 3)$$

Summary 2

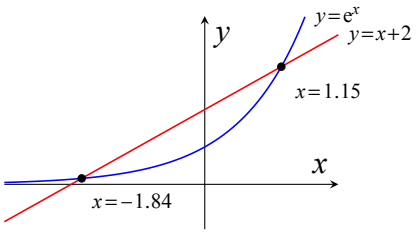
Inequalities

2.1 Solving Inequalities Graphically

- Rearrange to $f(x)>0$, or compare $y=f(x)$ with $y=g(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the x-values where the required graph is above or below.

Example

$$e^x < x+2$$



Using GC, intersections at $x=-1.8414$, $x=1.1462$

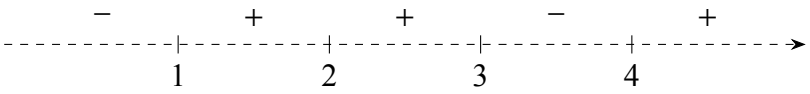
$$\therefore\ -1.84<x<1.15$$

2.2 Sign Test Technique

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant <0 and positive leading coefficient: $x^2+2x+5=(x+1)^2+4>0$.

Example

$$\frac{(x-1)(x-2)^2}{(x-3)(x-4)} \leqslant 0$$



$x=2$: numerator $=0$ (included)

$x=3,\ x=4$: denominator $=0$ (excluded)

$\therefore x\leqslant 1$ or $x=2$ or $3<x<4$

Note

Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the sign the same.

2.3 Common Substitutions

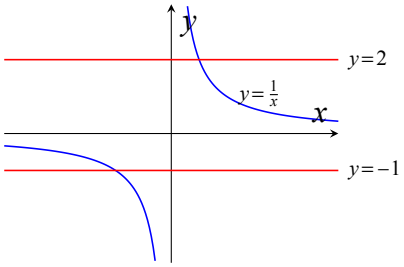
Replace x by	Original inequality		
	$x>2$	$x<-1$	$-1<x<2$
e^x	$x>\ln 2$	no solution	$x<\ln 2$
$\ln x$	$x>e^2$	$0<x<e^{-1}$	$e^{-1}<x<e^2$
$-x$	$x<-2$	$x>1$	$-2<x<1$
$x+1$	$x>1$	$x<-2$	$-2<x<1$
$\lvert x\rvert$	$x<-2$ or $x>2$	no solution	$-2<x<2$
x^2	$x<-\sqrt{2}$ or $x>\sqrt{2}$	no solution	$-\sqrt{2}<x<\sqrt{2}$
$\frac{1}{x}$	$\left(0<x<\frac{1}{2}\right)$	$(-1<x<0)$	$\left(x<-1 \text{ or } x>\frac{1}{2}\right)$
$\cos x,\ 0\leqslant x\leqslant 2\pi$	(no solution)	(no solution)	$(0\leqslant x<\pi \text{ or } \pi<x\leqslant 2\pi)$

Regardless of whatever substitution you use, using a graph will always work, but some can be solved without a graph.

For $\frac{1}{x}$, $\sin x$ and $\cos x$, draw the graph to solve.

Example

$$\frac{1}{x}>2, \quad \frac{1}{x}<-1, \quad -1<\frac{1}{x}<2$$



Example

$$\cos x>\tfrac{1}{2}, \quad \cos x<-\tfrac{1}{2},$$
$$-\tfrac{1}{2}<\cos x<\tfrac{1}{2}, \quad 0\leqslant x\leqslant 2\pi$$

